



AZ-104

MICROSOFT AZURE

COURSE SLIDES · WSQ

Microsoft Certified: Azure Administrator Associate (AZ-104)

WSQ Course Code: TGS-2023039182

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COURSE ADMINISTRATION

Welcome & Housekeeping

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

CERTIFICATION

Microsoft Azure certified — cloud infrastructure, DevOps & automation.

DELIVERS

WSQ courses on Azure, cloud administration & software engineering.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

- Your name and organisation / role.
- Your experience with Azure or other clouds (if any).
- What you want to build or administer on Azure after this course.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

5

Be punctual; return from breaks on time.

6

75% attendance is required.

LMS / TMS

- Access your course materials, attendance and assessment on the LMS/TMS portal.
- Portal: <https://lms-tms.tertiaryinfotech.com>
- Download the slides and Learner Guide for reference during the open-book assessment.

SCHEDULE

Lesson Plan — 3 Days, 8 hours/day

Days 1–2

- Day 1 — Identities, Governance & Storage
 - Topic 1: Identities & Governance (Labs 1–6)
 - Topic 2: Storage (Labs 7–11)
- Day 2 — Compute & Virtual Networking
 - Topic 3: Compute (Labs 12–17)
 - Topic 4: Networking begins (Labs 18–19)

Day 3 & timing

- Day 3 — Networking, Monitoring, Maintenance & Assessment
 - Topic 4: Networking (Labs 20–22)
 - Topic 5: Monitoring & Maintenance (Labs 23–26)
 - Final Assessment (WA + PP)
- Daily timing
 - 9:00am–6:00pm · 1-hour lunch · tea breaks within

Learning Outcomes

1

Identities & Governance

Entra users/groups, RBAC, Policy, resource groups, locks, tags, cost.

2

Storage

Storage accounts, redundancy, access control, Blob, Files, data movement.

3

Compute

ARM/Bicep, virtual machines, scale sets, containers, App Service.

4

Virtual Networking

VNets, NSGs, Bastion, endpoints, DNS, load balancing.

5

Monitor & Maintain

Azure Monitor, Network Watcher, Azure Backup, Site Recovery.

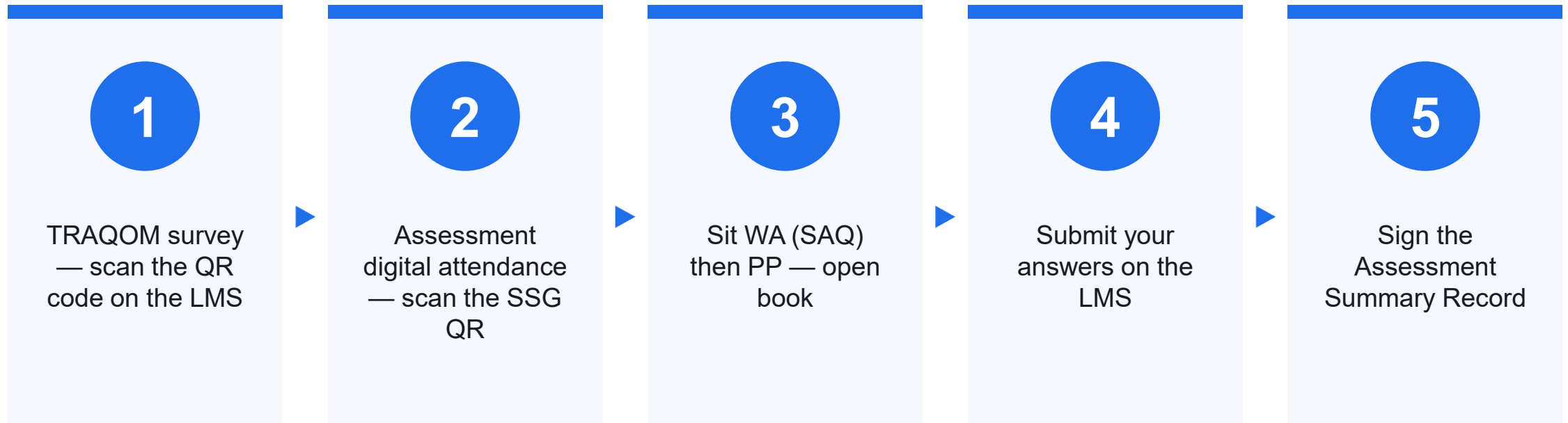
Briefing for Assessment

- Place phones and other materials under the table or on the floor.
- No photos or recording of assessment scripts.
- No discussion during the assessment.
- Use a black/blue pen for hard-copy assessments.
- No liquid paper / correction tape.
- Scripts are collected when time is up.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on Azure tasks, 1 hour, open book.
- Format: Open Book — slides, Learner Guide and approved materials only.
- A minimum of 75% attendance is required to be eligible for assessment and funding.
- An appeal process is available if required.

Assessment Flow





CORE CONCEPTS

Azure Fundamentals for Administrators

What is Microsoft Azure?

1

Public cloud

Compute, storage, networking, identity and platform services on demand.

2

Pay-as-you-go

Rent capacity instead of buying and running your own hardware.

3

Many ways to manage

Azure Portal, Azure CLI, Azure PowerShell and ARM/Bicep templates.

4

The AZ-104 role

Implement, manage and monitor an organisation's Azure environment.

The Azure Resource Hierarchy

Top of the tree

- Microsoft Entra tenant (identity)
- Management groups
- Group many subscriptions

Billing & access

- Subscription = billing boundary
- Resource groups = lifecycle folder
- Delete the RG, delete its contents

Governance flows down

- Policy, RBAC, locks, tags, budgets
- Applied at a scope
- Inherited by everything below

WHY IT MATTERS

Everything is a resource in a resource group.

Master the hierarchy — scopes are where identity, governance, cost and access all come together.

Three ways to administer Azure

Interactive

- Azure Portal
 - Visual, discover features and blades
 - Best when learning a task
- Azure CLI (az)
 - Cross-platform commands in Cloud Shell
 - Fast, scriptable

Command-line & code

- Azure PowerShell (Az)
 - Cmdlets for Windows-centric teams
 - Same result as CLI
- ARM / Bicep
 - Infrastructure as code
 - Repeatable deployments

Azure Cloud Shell

- Browser-based shell at <https://shell.azure.com> — nothing to install locally.
- Pre-loaded with the az CLI, Az PowerShell, azcopy and Bicep.
- Bash for az commands; PowerShell for Az cmdlets.
- Every lab in this course runs in the Portal and Cloud Shell.

TOPIC 01

Manage Azure Identities and Governance

Microsoft Entra ID · RBAC · Azure Policy · Subscriptions · Cost

Key Concepts — Manage Azure Identities and Governance

1

Microsoft Entra ID authenticates identities (users, groups, guests); Azure RBAC authorises what they may do.

2

An RBAC role assignment = security principal + role definition + scope (management group / subscription / RG / resource).

3

Governance is applied at a scope and inherited downward: Azure Policy, resource locks, tags, budgets.

4

The resource hierarchy: management group → subscription → resource group → resource.

Hands-On Labs — Manage Azure Identities and Governance

Labs 1–2

- Manage Microsoft Entra Users and Groups
- Manage External (B2B Guest) Users and Configure SSPR

Labs 3–4

- Manage Access to Azure Resources with RBAC
- Implement and Manage Azure Policy

Labs 5–6

- Manage Resource Groups, Locks and Tags
- Manage Costs and Configure Management Groups

Manage Microsoft Entra Users and Groups

LAB 1

The learner creates cloud users, builds security and Microsoft 365 groups, manages membership and properties, and assigns licenses in Microsoft Entra ID.

You'll build

A team of Entra users, a security group, a Microsoft 365 group, and a dynamic group

Azure services: Microsoft Entra ID, az ad, Microsoft Graph, licenses

Manage Microsoft Entra Users and Groups

STEP 1 OF 8



Sign in and confirm your tenant primary domain

```
$ DOMAIN=$(az rest --method get --url "https://graph.microsoft.com/v1.0/domains?$select=id,isDefault" --query "value[?isDefault].id" -o tsv)
```


Manage Microsoft Entra Users and Groups

STEP 2 OF 8

2

Create individual cloud users

```
$ az ad user create --display-name "Jane Doe" --user-principal-name "jdoe@$DOMAIN" --password "P@ssw0rd-Lab1!" --force-change-password-next-sign-in true
```

Manage Microsoft Entra Users and Groups

STEP 3 OF 8

3

Edit user job title, department and usage location

```
$ az ad user update --id "jdoe@$DOMAIN" --set jobTitle="Cloud Administrator" department="IT" usageLocation="SG"
```

Manage Microsoft Entra Users and Groups

STEP 4 OF 8

4

Bulk create users from a list (Portal bulk create)

```
$ az ad user create --display-name "Amy Tan" --user-principal-name "atan@$DOMAIN" --password "P@ssw0rd-Lab1!" --force-change-password-next-sign-in true
```

Manage Microsoft Entra Users and Groups

STEP 5 OF 8

5

Create a security group and a Microsoft 365 group

```
$ az ad group create --display-name "grp-az104-admins" --mail-nickname "grp-az104-admins"
```

Manage Microsoft Entra Users and Groups

STEP 6 OF 8



6

Add users as members and verify membership

```
$ az ad group member add --group "$GROUP_ID" --member-id "$JANE_ID"
```

Manage Microsoft Entra Users and Groups

STEP 7 OF 8

7

Create a dynamic-membership group driven by department

```
$ az rest --method post --url "https://graph.microsoft.com/v1.0/groups" --headers "Content-Type=application/json" --body '{"displayName":"grp-az104-it-dynamic","mailNickname":"grp-az104-it-dynamic","mailEnabled":false,"securityEnabled":true,"groupTypes":["DynamicMembership"],"membershipRule":"(user.department -eq \"IT\")","membershipRuleProcessingState\":\"On\"}'
```

Manage Microsoft Entra Users and Groups

STEP 8 OF 8

8

Inspect tenant SKUs and assign a license to a user

```
$ az rest --method post --url "https://graph.microsoft.com/v1.0/users/jdoe@$DOMAIN/assignLicense" --headers "Content-Type=application/json" --body '{ "addLicenses": [ { "skuId": "<SKU_ID>", "disabledPlans": [] } ], "removeLicenses": [] }'
```

Manage Microsoft Entra Users and Groups

Test it

Verify the users appear in All users, the groups list their members, the dynamic rule auto-adds IT users, and the assigned license shows on the user.

Manage External (B2B Guest) Users and Configure SSPR

LAB 2

The learner invites a B2B guest user, reviews external collaboration and cross-tenant settings, then enables and scopes self-service password reset with authentication methods and registration.

You'll build

A redeemed B2B guest user, a partners group, and a scoped SSPR configuration

Azure services: Microsoft Entra ID, External Identities / B2B, Password reset (SSPR), az rest

Manage External (B2B Guest) Users and Configure SSPR

STEP 1 OF 8

1

Sign in and set the tenant primary-domain variable

```
$ DOMAIN=$(az rest --method get --url "https://graph.microsoft.com/v1.0/domains?$select=id,isDefault" --query "value[?isDefault].id" -o tsv)
```

Manage External (B2B Guest) Users and Configure SSPR

STEP 2 OF 8

2

Invite a B2B guest user by email

```
$ az rest --method post --url "https://graph.microsoft.com/v1.0/invitations" --headers "Content-Type=application/json" --body '{ "invitedUserEmailAddress": "partner.test@gmail.com", "invitedUserDisplayName": "Contoso Partner", "inviteRedirectUrl": "https://myapps.microsoft.com", "sendInvitationMessage": true }'
```

Manage External (B2B Guest) Users and Configure SSPR

STEP 3 OF 8

3

Verify the guest and add it to a partners group

```
$ az ad group member add --group "$GRP_ID" --member-id "$GUEST_ID"
```

Manage External (B2B Guest) Users and Configure SSPR

STEP 4 OF 8

4

Review External collaboration settings and authorization policy

```
$ az rest --method get --url "https://graph.microsoft.com/v1.0/policies/authorizationPolicy" --query "{allowInvitesFrom:allowInvitesFrom, guestUserRole:guestRoleId}"
```

Manage External (B2B Guest) Users and Configure SSPR

STEP 5 OF 8



5

Review Cross-tenant access (B2B) default settings

Manage External (B2B Guest) Users and Configure SSPR

STEP 6 OF 8



6

Enable and scope SSPR to a selected group (Portal)

Manage External (B2B Guest) Users and Configure SSPR

STEP 7 OF 8



Configure SSPR authentication methods, registration and notifications (Portal)

Manage External (B2B Guest) Users and Configure SSPR

STEP 8 OF 8

8

Test the reset flow; contrast SSPR with an admin reset

```
$ az ad user update --id "jdoe@$DOMAIN" --password "N3wP@ssw0rd-Lab2!" --force-change-password-next-sign-in true
```

Manage External (B2B Guest) Users and Configure SSPR

Test it

Verify the guest shows User type = Guest with an #EXT# UPN, and a scoped test user can register methods and reset their password via <https://aka.ms/sspr>.

Manage Access to Azure Resources with RBAC

LAB 3

The learner assigns built-in Azure roles at subscription, resource-group and resource scopes, interprets effective access with Check access, then builds and assigns a custom role.

You'll build

Role assignments at three scopes plus a custom 'VM Restart Operator' role

Azure services: Azure RBAC (role assignments, role definitions), Resource Manager, Storage account

Manage Access to Azure Resources with RBAC

STEP 1 OF 8

1

Set variables and create a target resource group

```
$ az group create --name "rg-az104-lab03" --location "southeastasia"
```

Manage Access to Azure Resources with RBAC

STEP 2 OF 8

2

Create a storage account to act as a scoped resource

```
$ az storage account create --name "$SA" --resource-group "$RG" --location "$LOCATION" --sku Standard_LRS
```

Manage Access to Azure Resources with RBAC

STEP 3 OF 8

3

Explore built-in role definitions (Owner/Contributor/Reader)

```
$ az role definition list --name "Contributor" --query "[0].permissions[0].{actions:actions, notActions:notActions}"
```

Manage Access to Azure Resources with RBAC

STEP 4 OF 8

4

Assign Reader to a group at subscription scope

```
$ az role assignment create --assignee-object-id "$GRP_ID" --assignee-principal-type Group --role "Reader" --scope "/subscriptions/$SUB_ID"
```

Manage Access to Azure Resources with RBAC

STEP 5 OF 8

5

Assign Contributor at resource-group scope

```
$ az role assignment create --assignee-object-id "$JANE_ID" --assignee-principal-type User --role "Contributor" --scope "/subscriptions/$SUB_ID/resourceGroups/$RG"
```


Manage Access to Azure Resources with RBAC

STEP 6 OF 8

6

Assign Storage Blob Data Reader at resource scope

```
$ az role assignment create --assignee-object-id "$RAJ_ID" --assignee-principal-type User --role "Storage Blob Data Reader" --scope "$SA_ID"
```

Manage Access to Azure Resources with RBAC

STEP 7 OF 8



Interpret effective access including inherited assignments

```
$ az role assignment list --scope "$SA_ID" --include-inherited -o table
```

Manage Access to Azure Resources with RBAC

STEP 8 OF 8



8

Create and assign a minimal custom role

```
$ az role definition create --role-definition /tmp/vm-restart-role.json
```

Manage Access to Azure Resources with RBAC

Test it

Use Check access and 'az role assignment list --include-inherited' to confirm each principal has the expected role at the right scope, and that the custom role is assignable.

Implement and Manage Azure Policy

LAB 4

The learner assigns a built-in policy, authors a custom definition, bundles policies into an initiative, reads compliance, and remediates non-compliant resources with a managed identity.

You'll build

An Allowed-locations assignment, a custom tag policy, a governance initiative, and a remediation task

Azure services: Azure Policy (definitions, initiatives, assignments, remediation), managed identity, Storage accounts

Implement and Manage Azure Policy

STEP 1 OF 8

1

Create a resource group as the policy scope

```
$ az group create --name "rg-az104-lab04" --location "southeastasia"
```

Implement and Manage Azure Policy

STEP 2 OF 8

2

Assign the built-in Allowed locations policy

```
$ az policy assignment create --name "allowed-locations-lab04" --policy "$LOC_DEF" --scope "$SCOPE" --params '{  
  "listOfAllowedLocations": { "value": [ "southeastasia" ] } }'
```

Implement and Manage Azure Policy

STEP 3 OF 8

3

Test the Deny effect against a disallowed region

```
$ az storage account create --name "stlab04deny$RANDOM" --resource-group "$RG" --location "eastus" --sku Standard_LRS
```


Implement and Manage Azure Policy

STEP 4 OF 8

4

Author a custom policy definition requiring a tag

```
$ az policy definition create --name "require-env-tag-on-storage" --display-name "Storage accounts must have an Environment tag" --mode Indexed --rules /tmp/require-env-tag.json --subscription "$SUB_ID"
```

Implement and Manage Azure Policy

STEP 5 OF 8

5

Create an initiative (policy set) and assign it

```
$ az policy set-definition create --name "az104-governance-baseline" --display-name "AZ-104 Governance Baseline" --definitions /tmp/initiative.json --subscription "$SUB_ID"
```

Implement and Manage Azure Policy

STEP 6 OF 8

6

Trigger a scan and interpret compliance state

```
$ az policy state list --resource-group "$RG" --query "[].{resource:resourceId, state:complianceState, policy:policyDefinitionName}" -o table
```

Implement and Manage Azure Policy

STEP 7 OF 8



Assign a Modify policy with managed identity and remediate

```
$ az policy remediation create --name "remediate-env-tag" --resource-group "$RG" --policy-assignment "$ASSIGN_ID"
```

Implement and Manage Azure Policy

STEP 8 OF 8



8

Add an exemption to waive a specific resource

```
$ az policy exemption create --name "exempt-legacy-sa" --policy-assignment "$ASSIGN_ID" --exemption-category  
Waiver --scope "$SA_ID" --description "Legacy account exempt from Environment tag until migrated"
```

Implement and Manage Azure Policy

Test it

Confirm the Deny effect blocks the disallowed region, the Compliance blade shows compliant/non-compliant counts, and the remediation task tags existing resources.

Manage Resource Groups, Locks and Tags

LAB 5

The learner creates resource groups, deploys a resource, applies and updates tags, moves the resource between groups, and protects resources with CanNotDelete and ReadOnly locks.

You'll build

Two resource groups, a tagged storage account moved to an archive RG, and delete/read-only locks

Azure services: Azure Resource Manager (resource groups, tags, resource locks, resource move), Storage account

Manage Resource Groups, Locks and Tags

STEP 1 OF 8



Create two resource groups

```
$ az group create --name "rg-az104-lab05" --location "southeastasia"
```


Manage Resource Groups, Locks and Tags

STEP 2 OF 8

2

Deploy a storage account to tag, move and lock

```
$ az storage account create --name "$SA" --resource-group "$RG" --location "$LOCATION" --sku Standard_LRS
```

Manage Resource Groups, Locks and Tags

STEP 3 OF 8



3

Apply tags to the resource group

```
$ az tag create --resource-id "/subscriptions/$(az account show --query id -o tsv)/resourceGroups/$RG" --tags  
CostCenter=1001 Environment=Dev Owner=angch
```

Manage Resource Groups, Locks and Tags

STEP 4 OF 8

4

Merge, update and delete tags on the resource

```
$ az tag update --resource-id "$SA_ID" --operation Merge --tags Environment=Dev Project=AZ104
```

Manage Resource Groups, Locks and Tags

STEP 5 OF 8

5

Move the resource to the archive resource group

```
$ az resource move --destination-group "$RG2" --ids "$SA_ID"
```

Manage Resource Groups, Locks and Tags

STEP 6 OF 8

6

Apply a CanNotDelete lock on the archive RG

```
$ az lock create --name "lock-no-delete" --lock-type CanNotDelete --resource-group "$RG2" --notes "Protect archived resources from deletion"
```

Manage Resource Groups, Locks and Tags

STEP 7 OF 8

7

Test the lock, then apply a ReadOnly lock on the resource

```
$ az lock create --name "lock-readonly" --lock-type ReadOnly --resource-group "$RG2" --resource-name "$SA" --resource-type "Microsoft.Storage/storageAccounts" --namespace "Microsoft.Storage"
```

Manage Resource Groups, Locks and Tags

STEP 8 OF 8



8

Manage subscription-level resource providers

```
$ az provider register --namespace "Microsoft.Storage"
```

Manage Resource Groups, Locks and Tags

Test it

Verify the tags read back correctly, the resource now lives in the archive RG, and a delete attempt fails with a ScopeLocked error while the locks are in place.

Manage Costs and Configure Management Groups

LAB 6

The learner analyzes costs, creates a budget with alert thresholds, reviews Azure Advisor cost recommendations, and builds a management-group hierarchy to apply governance at scale.

You'll build

A monthly budget with alerts, an action group, and an mg-az104 management-group hierarchy

Azure services: Microsoft Cost Management (Cost analysis, budgets, alerts), Azure Advisor, Management groups

Manage Costs and Configure Management Groups

STEP 1 OF 8

1

Set subscription and tenant variables

```
$ SUB_ID=$(az account show --query id -o tsv)
```

Manage Costs and Configure Management Groups

STEP 2 OF 8

2

Analyze month-to-date cost by service

```
$ az costmanagement query --type ActualCost --scope "/subscriptions/$SUB_ID" --timeframe MonthToDate --dataset-aggregation '{ "totalCost": { "name": "Cost", "function": "Sum" } }' --dataset-grouping name="ServiceName" type="Dimension" -o table
```

Manage Costs and Configure Management Groups

STEP 3 OF 8



3

Create an action group for budget alert emails

```
$ az monitor action-group create --name "ag-budget-alerts" --resource-group "$RG_MON" --short-name "budgetAG" --  
action email admin angch@tertiaryinfotech.com
```

Manage Costs and Configure Management Groups

STEP 4 OF 8

4

Create a monthly budget with alert thresholds

```
$ az consumption budget create --budget-name "budget-az104-monthly" --amount 100 --category Cost --time-grain Monthly --start-date "2026-07-01" --end-date "2027-06-30" --notifications '{ "warning80": { "enabled": true, "operator": "GreaterThanOrEqualTo", "threshold": 80, "contactEmails": ["angch@tertiaryinfotech.com"] } }'
```

Manage Costs and Configure Management Groups

STEP 5 OF 8

5

Review fired cost / anomaly alerts

```
$ az rest --method get --url  
"https://management.azure.com/subscriptions/$SUB_ID/providers/Microsoft.CostManagement/alerts?api-  
version=2023-08-01" --query "value[].{name:name, type:properties.definition.type, status:properties.status}" -o table
```

Manage Costs and Configure Management Groups

STEP 6 OF 8

6

Review Azure Advisor cost recommendations

```
$ az advisor recommendation list --query "[?category=='Cost'].{impact:impact, problem:shortDescription.problem, resource:resourceMetadata.resourceId}" -o table
```

Manage Costs and Configure Management Groups

STEP 7 OF 8



Create a management-group hierarchy

```
$ az account management-group create --name "mg-az104-prod" --display-name "AZ104 Prod" --parent "mg-az104-root"
```


Manage Costs and Configure Management Groups

STEP 8 OF 8



8

Move the subscription and assign inherited governance

```
$ az account management-group subscription add --name "mg-az104-nonprod" --subscription "$SUB_ID"
```

Manage Costs and Configure Management Groups

Test it

Confirm the budget and action group appear with the configured thresholds, Advisor lists cost recommendations, and the subscription sits under the management-group hierarchy with the role inherited downward.

Recap — Manage Azure Identities and Governance

- You can now: Manage Microsoft Entra users and groups
- You can now: Manage external users and self-service password reset
- You can now: Manage access to Azure resources
- You can now: Implement and manage Azure Policy
- You can now: Manage resource groups and subscriptions
- You can now: Manage subscriptions and governance

TOPIC 02

Implement and Manage Storage

Storage accounts · Redundancy · Access · Blob · Files · AzCopy

Key Concepts — Implement and Manage Storage

1

A storage account is a globally-unique namespace holding Blobs, Files, Queues and Tables.

2

Redundancy (LRS / ZRS / GRS / RA-GRS) trades cost against durability and regional failover.

3

Access is controlled by keys, SAS tokens, Entra RBAC and storage firewalls / VNet rules.

4

Blob access tiers (hot / cool / cold / archive), versioning, soft delete and lifecycle rules manage cost.

Hands-On Labs — Implement and Manage Storage

Labs 7–8

- Create and Configure Storage Accounts and Redundancy
- Configure Access to Storage: Firewalls, SAS, Access Keys

Labs 9–10

- Configure Azure Blob Storage: Tiers, Lifecycle, Versioning
- Configure Azure Files and Identity-Based Access

Labs 11–11

- Manage Data with Storage Explorer and AzCopy

Create and Configure Storage Accounts and Redundancy

LAB 7

The learner creates general-purpose v2 storage accounts, switches redundancy tiers, configures encryption, and sets up object replication across accounts.

You'll build

A source and destination storage account with an object replication policy.

Azure services: Azure Storage, Blob storage, Azure Key Vault, object replication, az storage account

Create and Configure Storage Accounts and Redundancy

STEP 1 OF 8



Create a resource group

```
$ az group create --name rg-az104-lab07 --location eastus
```


Create and Configure Storage Accounts and Redundancy

STEP 2 OF 8

2

Create an LRS storage account

```
$ az storage account create --name $SRC --resource-group rg-az104-lab07 --location eastus --sku Standard_LRS --kind StorageV2 --access-tier Hot
```

Create and Configure Storage Accounts and Redundancy

STEP 3 OF 8

3

Change redundancy to RA-GRS

```
$ az storage account update --name $SRC --resource-group rg-az104-lab07 --sku Standard_RAGRS
```

Create and Configure Storage Accounts and Redundancy

STEP 4 OF 8

4

Inspect and set customer-managed key encryption

```
$ az storage account update --name $SRC --resource-group rg-az104-lab07 --encryption-key-source  
Microsoft.Keyvault --encryption-key-vault "https://$KV.vault.azure.net/" --encryption-key-name storagecmk
```

Create and Configure Storage Accounts and Redundancy

STEP 5 OF 8

5

Create a destination account for replication

```
$ az storage account create --name $DST --resource-group rg-az104-lab07 --location westus --sku Standard_LRS --kind StorageV2
```

Create and Configure Storage Accounts and Redundancy

STEP 6 OF 8

6

Enable versioning and change feed on both

```
$ az storage account blob-service-properties update --account-name $ACC --resource-group rg-az104-lab07 --enable-versioning true --enable-change-feed true
```

Create and Configure Storage Accounts and Redundancy

STEP 7 OF 8



Create containers and upload a blob

```
$ az storage blob upload --account-name $SRC --container-name orders --name sample.txt --file sample.txt --auth-mode login
```

Create and Configure Storage Accounts and Redundancy

STEP 8 OF 8

8

Configure an object replication policy

```
$ az storage account or-policy create --account-name $DST --resource-group rg-az104-lab07 --source-account $SRC  
--destination-account $DST --source-container orders --destination-container orders --min-creation-time '2024-01-  
01T00:00:00Z'
```

Create and Configure Storage Accounts and Redundancy

Test it

List blobs on the destination account and confirm the source blob has replicated asynchronously.

Configure Access to Storage: Firewalls, SAS, Access Keys

LAB 8

The learner rotates access keys, generates account and service SAS tokens, binds a SAS to a stored access policy, locks the account behind a firewall, and grants RBAC data access.

You'll build

A firewalled storage account secured with SAS tokens, stored access policies, and RBAC roles.

Azure services: Azure Storage, storage firewall + virtual networks, SAS, stored access policies, access keys, Azure RBAC

Configure Access to Storage: Firewalls, SAS, Access Keys

STEP 1 OF 8

1

Create the resource group and storage account

```
$ az storage account create --name $SA --resource-group rg-az104-lab08 --location eastus --sku Standard_LRS --kind StorageV2
```

Configure Access to Storage: Firewalls, SAS, Access Keys

STEP 2 OF 8

2

Create a container and upload a blob

```
$ az storage blob upload --account-name $SA --container-name docs --name report.txt --file report.txt --auth-mode login
```

Configure Access to Storage: Firewalls, SAS, Access Keys

STEP 3 OF 8

3

List and rotate access keys

```
$ az storage account keys renew --account-name $SA --resource-group rg-az104-lab08 --key primary
```

Configure Access to Storage: Firewalls, SAS, Access Keys

STEP 4 OF 8

4

Generate an account SAS

```
$ az storage account generate-sas --account-name $SA --account-key $KEY --services b --resource-types sco --permissions rl --expiry $EXPIRY --https-only -o tsv
```

Configure Access to Storage: Firewalls, SAS, Access Keys

STEP 5 OF 8

5

Generate a service SAS for one blob

```
$ az storage blob generate-sas --account-name $SA --account-key $KEY --container-name docs --name report.txt --permissions r --expiry $EXPIRY --https-only --full-uri -o tsv
```

Configure Access to Storage: Firewalls, SAS, Access Keys

STEP 6 OF 8

6

Create a stored access policy and bind a SAS

```
$ az storage container policy create --account-name $SA --account-key $KEY --container-name docs --name readonly  
--permissions rl --expiry $POLICYEXP
```

Configure Access to Storage: Firewalls, SAS, Access Keys

STEP 7 OF 8



Configure the storage firewall and VNet rules

```
$ az storage account update --name $SA --resource-group rg-az104-lab08 --default-action Deny --bypass  
AzureServices
```


Configure Access to Storage: Firewalls, SAS, Access Keys

STEP 8 OF 8

8

Grant identity-based RBAC access

```
$ az role assignment create --assignee $USER --role "Storage Blob Data Contributor" --scope $SCOPE
```

Configure Access to Storage: Firewalls, SAS, Access Keys

Test it

List blobs with `--auth-mode login` using only your RBAC identity, with no key or SAS supplied.

Configure Azure Blob Storage: Tiers, Lifecycle, Versioning

LAB 9

The learner creates a blob container, moves blobs between hot/cool/cold/archive tiers, enables versioning and soft delete, and authors a JSON lifecycle management policy.

You'll build

A blob container with tiered blobs, soft delete, versioning, and an automated lifecycle policy.

Azure services: Azure Blob Storage, access tiers, blob/container soft delete, blob versioning, lifecycle management

Configure Azure Blob Storage: Tiers, Lifecycle, Versioning

STEP 1 OF 8

1

Create the resource group and storage account

```
$ az storage account create --name $SA --resource-group rg-az104-lab09 --location eastus --sku Standard_LRS --kind StorageV2 --access-tier Hot
```

Configure Azure Blob Storage: Tiers, Lifecycle, Versioning

STEP 2 OF 8

2

Create a blob container

```
$ az storage container create --account-name $SA --name media --auth-mode login
```

Configure Azure Blob Storage: Tiers, Lifecycle, Versioning

STEP 3 OF 8

3

Upload blobs into different tiers

```
$ az storage blob upload --account-name $SA --container-name media --name hot.txt --file hot.txt --tier Hot --auth-mode login
```

Configure Azure Blob Storage: Tiers, Lifecycle, Versioning

STEP 4 OF 8

4

Change the access tier of a blob

```
$ az storage blob set-tier --account-name $SA --container-name media --name hot.txt --tier Archive --auth-mode login
```

Configure Azure Blob Storage: Tiers, Lifecycle, Versioning

STEP 5 OF 8

5

Enable blob versioning

```
$ az storage account blob-service-properties update --account-name $SA --resource-group rg-az104-lab09 --enable-versioning true
```


Configure Azure Blob Storage: Tiers, Lifecycle, Versioning

STEP 6 OF 8

6

Enable soft delete for blobs and containers

```
$ az storage account blob-service-properties update --account-name $SA --resource-group rg-az104-lab09 --enable-delete-retention true --delete-retention-days 7
```

Configure Azure Blob Storage: Tiers, Lifecycle, Versioning

STEP 7 OF 8



Author a JSON lifecycle management policy

```
$ az storage account management-policy create --account-name $SA --resource-group rg-az104-lab09 --policy @policy.json
```

Configure Azure Blob Storage: Tiers, Lifecycle, Versioning

STEP 8 OF 8



8

Verify the lifecycle policy

```
$ az storage account management-policy show --account-name $SA --resource-group rg-az104-lab09 -o json
```

Configure Azure Blob Storage: Tiers, Lifecycle, Versioning

Test it

Show the management policy and confirm the tier-down and expire rules match the JSON you applied.

Configure Azure Files and Identity-Based Access

LAB 10

The learner creates an SMB file share, uploads files, takes and restores snapshots, enables share soft delete, and configures identity-based authentication with share-level RBAC.

You'll build

An SMB file share with snapshots, soft delete, and identity-based access.

Azure services: Azure Files, share snapshots, file-share soft delete, identity-based authentication, Azure RBAC share roles

Configure Azure Files and Identity-Based Access

STEP 1 OF 8

1

Create the resource group and storage account

```
$ az storage account create --name $SA --resource-group rg-az104-lab10 --location eastus --sku Standard_LRS --kind StorageV2
```

Configure Azure Files and Identity-Based Access

STEP 2 OF 8

2

Create a file share with a quota

```
$ az storage share-rm create --resource-group rg-az104-lab10 --storage-account $SA --name projects --quota 100 --access-tier TransactionOptimized
```

Configure Azure Files and Identity-Based Access

STEP 3 OF 8

3

Upload a file and list the share

```
$ az storage file upload --account-name $SA --account-key $KEY --share-name projects --source plan.txt
```


Configure Azure Files and Identity-Based Access

STEP 4 OF 8

4

Enable share soft delete

```
$ az storage account file-service-properties update --account-name $SA --resource-group rg-az104-lab10 --enable-delete-retention true --delete-retention-days 7
```

Configure Azure Files and Identity-Based Access

STEP 5 OF 8

5

Take a share snapshot

```
$ az storage share snapshot --account-name $SA --account-key $KEY --name projects --query snapshot -o tsv
```

Configure Azure Files and Identity-Based Access

STEP 6 OF 8



6

Restore a file from a snapshot

```
$ az storage file copy start --account-name $SA --account-key $KEY --source-share projects --source-path plan.txt --source-snapshot "$SNAP" --destination-share projects --destination-path plan-restored.txt
```

Configure Azure Files and Identity-Based Access

STEP 7 OF 8



Configure identity-based authentication

```
$ az storage account update --name $SA --resource-group rg-az104-lab10 --enable-files-aadds true
```

Configure Azure Files and Identity-Based Access

STEP 8 OF 8

8

Assign a share-level RBAC role

```
$ az role assignment create --assignee $USER --role "Storage File Data SMB Share Contributor" --scope "$SHARESCOPE"
```

Configure Azure Files and Identity-Based Access

Test it

List files inside the snapshot and confirm the restored file appears back in the live share.

Manage Data with Storage Explorer and AzCopy

LAB 11

The learner provisions an account with two containers, then moves data with Azure Storage Explorer, AzCopy via SAS, and AzCopy via Entra login including a server-side copy.

You'll build

A storage account with data transferred via Storage Explorer and AzCopy.

Azure services: Azure Storage (Blob), Azure Storage Explorer, AzCopy, SAS, Azure RBAC

Manage Data with Storage Explorer and AzCopy

STEP 1 OF 8

1

Create the account and two containers

```
$ az storage container create --account-name $SA --name source --auth-mode login
```


Manage Data with Storage Explorer and AzCopy

STEP 2 OF 8



2

Grant yourself data-plane RBAC

```
$ az role assignment create --assignee $USER --role "Storage Blob Data Contributor" --scope $SCOPE
```

Manage Data with Storage Explorer and AzCopy

STEP 3 OF 8



Install and connect Azure Storage Explorer

Manage Data with Storage Explorer and AzCopy

STEP 4 OF 8



Upload and download with the GUI

Manage Data with Storage Explorer and AzCopy

STEP 5 OF 8

5

Create test data and a container SAS

```
$ az storage container generate-sas --account-name $SA --account-key $KEY --name source --permissions racwdl --  
expiry $EXPIRY --https-only -o tsv
```

Manage Data with Storage Explorer and AzCopy

STEP 6 OF 8

6

Upload a folder with AzCopy using a SAS

```
$ azcopy copy "$HOME/azcopydemo/*" "https://$SA.blob.core.windows.net/source?$SAS" --recursive=true
```

Manage Data with Storage Explorer and AzCopy

STEP 7 OF 8



Copy account-to-account server-side

```
$ azcopy copy "https://$SA.blob.core.windows.net/source?$SAS"  
"https://$SA.blob.core.windows.net/backup?$BACKUPSAS" --recursive=true
```

Manage Data with Storage Explorer and AzCopy

STEP 8 OF 8

8

Use AzCopy with Entra login and sync

```
$ azcopy sync "$HOME/azcopydemo" "https://$SA.blob.core.windows.net/source" --recursive=true
```

Manage Data with Storage Explorer and AzCopy

Test it

Run azcopy list on the source container URL and confirm the uploaded and synced files are present.

Recap — Implement and Manage Storage

- You can now: Configure access to storage
- You can now: Configure Azure Blob Storage
- You can now: Configure Azure Files
- You can now: Configure Azure Blob Storage / Configure access to storage

TOPIC 03

Deploy and Manage Azure Compute Resources

ARM & Bicep · VMs · Scale Sets · Containers · App Service

Key Concepts — Deploy and Manage Azure Compute Resources

1

Infrastructure as code with ARM templates and Bicep makes deployments repeatable and reviewable.

2

Choose the least-effort compute: VM → VM Scale Set → containers (ACI / Container Apps) → App Service.

3

Availability zones and sets, managed disks and VM sizes determine resilience and performance.

4

App Service and Container Apps are fully-managed PaaS for web apps, APIs and microservices.

Hands-On Labs — Deploy and Manage Azure Compute Resources

Labs 12–13

- Automate Deployment with ARM Templates and Bicep
- Create a Virtual Machine, Encryption at Host, and Moving VMs

Labs 14–15

- VM Sizes, Disks, and Availability (Zones & Sets)
- Deploy and Configure a Virtual Machine Scale Set (VMSS) with Autoscale

Labs 16–17

- Containers: ACR, Container Instances, and Container Apps
- Azure App Service: Plans, Scaling, TLS, Custom Domains, Backup, Networking, Slots

Automate Deployment with ARM Templates and Bicep

LAB 12

Read, edit, and deploy resources with ARM templates and Bicep, then export and decompile between the two formats.

You'll build

A storage account deployed from both an ARM template and a Bicep file.

Azure services: ARM, Bicep, az deployment, Azure Storage, Cloud Shell

Automate Deployment with ARM Templates and Bicep

STEP 1 OF 9



Create the resource group

```
$ az group create --name rg-az104-lab12 --location eastus
```

Automate Deployment with ARM Templates and Bicep

STEP 2 OF 9



2

Interpret an ARM template's parameters/resources/outputs

Automate Deployment with ARM Templates and Bicep

STEP 3 OF 9



3

Modify the ARM template to add SKU and harden TLS

Automate Deployment with ARM Templates and Bicep

STEP 4 OF 9

4

Deploy the ARM template to the resource group

```
$ az deployment group create --resource-group rg-az104-lab12 --template-file storage.json --parameters storageAccountName=$SANAME skuName=Standard_LRS
```

Automate Deployment with ARM Templates and Bicep

STEP 5 OF 9



Author and edit a Bicep file

Automate Deployment with ARM Templates and Bicep

STEP 6 OF 9

6

Compile the Bicep file to ARM JSON

```
$ az bicep build --file storage.bicep
```

Automate Deployment with ARM Templates and Bicep

STEP 7 OF 9



Deploy the Bicep file

```
$ az deployment group create --resource-group rg-az104-lab12 --template-file storage.bicep --parameters storageAccountName=$SANAME2 skuName=Standard_ZRS
```

Automate Deployment with ARM Templates and Bicep

STEP 8 OF 9



8

Export the deployment as an ARM template

```
$ az group export --name rg-az104-lab12 > exported-rg.json
```

Automate Deployment with ARM Templates and Bicep

STEP 9 OF 9

9

Decompile the ARM template to Bicep

```
$ az bicep decompile --file exported-rg.json
```

Automate Deployment with ARM Templates and Bicep

Test it

Confirm both storage accounts exist and the exported template decompiles to Bicep.

Create a Virtual Machine, Encryption at Host, and Moving VMs

LAB 13

Deploy a Linux VM with encryption at host enabled, then move it across resource groups, subscriptions, and regions.

You'll build

A Linux VM with host-based encryption, moved between resource groups and regions.

Azure services: Azure Virtual Machines, Managed Disks, Azure Resource Mover, ARM

Create a Virtual Machine, Encryption at Host, and Moving VMs

STEP 1 OF 7

1

Create source and destination resource groups

```
$ az group create --name rg-az104-lab13 --location eastus
```

Create a Virtual Machine, Encryption at Host, and Moving VMs

STEP 2 OF 7

2

Register the encryption-at-host feature

```
$ az feature register --namespace Microsoft.Compute --name EncryptionAtHost
```

Create a Virtual Machine, Encryption at Host, and Moving VMs

STEP 3 OF 7

3

Create a VM with encryption at host

```
$ az vm create --resource-group rg-az104-lab13 --name vm-lab13 --image Ubuntu2204 --size Standard_B1s --admin-username azureuser --generate-ssh-keys --encryption-at-host --public-ip-sku Standard
```

Create a Virtual Machine, Encryption at Host, and Moving VMs

STEP 4 OF 7

4

Verify encryption at host on the VM

```
$ az vm show -g rg-az104-lab13 -n vm-lab13 --query securityProfile.encryptionAtHost
```

Create a Virtual Machine, Encryption at Host, and Moving VMs

STEP 5 OF 7

5

Move the VM to another resource group

```
$ az resource move --destination-group rg-az104-lab13-dest --ids $VMID
```

Create a Virtual Machine, Encryption at Host, and Moving VMs

STEP 6 OF 7

6

Move the VM to another subscription

```
$ az resource move --destination-group rg-az104-lab13-dest --destination-subscription-id $TARGET_SUB --ids $VMID
```

Create a Virtual Machine, Encryption at Host, and Moving VMs

STEP 7 OF 7

7

Move the VM to another region with Resource Mover

```
$ az resource-mover move-collection create --name mc-lab13 --resource-group rg-az104-lab13 --source-region eastus --target-region westus2 --location eastus2euap --identity type=SystemAssigned
```

Create a Virtual Machine, Encryption at Host, and Moving VMs

Test it

Confirm encryptionAtHost is true and the VM appears in the destination resource group.

VM Sizes, Disks, and Availability (Zones & Sets)

LAB 14

Resize a VM, attach and expand a managed data disk, and deploy VMs into an availability zone and an availability set.

You'll build

A resized VM with an expanded data disk plus zonal and availability-set VMs.

Azure services: Azure Virtual Machines, Managed Disks, Availability Zones, Availability Sets

VM Sizes, Disks, and Availability (Zones & Sets)

STEP 1 OF 8



Create the resource group and a base VM

```
$ az vm create --resource-group rg-az104-lab14 --name vm-lab14 --image Ubuntu2204 --size Standard_B1s --admin-username azureuser --generate-ssh-keys --public-ip-sku Standard
```

VM Sizes, Disks, and Availability (Zones & Sets)

STEP 2 OF 8

2

List available and resizable VM sizes

```
$ az vm list-vm-resize-options --resource-group rg-az104-lab14 --name vm-lab14 --output table
```

VM Sizes, Disks, and Availability (Zones & Sets)

STEP 3 OF 8

3

Resize the VM

```
$ az vm resize --resource-group rg-az104-lab14 --name vm-lab14 --size Standard_B2s
```

VM Sizes, Disks, and Availability (Zones & Sets)

STEP 4 OF 8

4

Create and attach a managed data disk

```
$ az vm disk attach --resource-group rg-az104-lab14 --vm-name vm-lab14 --name datadisk-lab14
```

VM Sizes, Disks, and Availability (Zones & Sets)

STEP 5 OF 8

5

Expand the data disk

```
$ az disk update --resource-group rg-az104-lab14 --name datadisk-lab14 --size-gb 64
```

VM Sizes, Disks, and Availability (Zones & Sets)

STEP 6 OF 8



6

Deploy a VM into an availability zone

```
$ az vm create --resource-group rg-az104-lab14 --name vm-zone1 --image Ubuntu2204 --size Standard_B1s --zone 1  
--admin-username azureuser --generate-ssh-keys --public-ip-sku Standard
```

VM Sizes, Disks, and Availability (Zones & Sets)

STEP 7 OF 8



Create an availability set

```
$ az vm availability-set create --resource-group rg-az104-lab14 --name avset-lab14 --platform-fault-domain-count 2 --platform-update-domain-count 5
```


VM Sizes, Disks, and Availability (Zones & Sets)

STEP 8 OF 8

8

Deploy VMs into the availability set

```
$ az vm create -g rg-az104-lab14 -n vm-set-1 --availability-set avset-lab14 --image Ubuntu2204 --size Standard_B1s --admin-username azureuser --generate-ssh-keys
```

VM Sizes, Disks, and Availability (Zones & Sets)

Test it

Confirm the resized VM, the 64 GiB disk, the zonal VM, and both availability-set VMs exist.

Deploy and Configure a Virtual Machine Scale Set (VMSS) with Autoscale

LAB 15

Create a VMSS behind a load balancer and attach autoscale rules that add and remove instances based on CPU usage.

You'll build

A Flexible VMSS with CPU-based scale-out and scale-in autoscale rules.

Azure services: Azure Virtual Machine Scale Sets, Azure Load Balancer, Azure Monitor Autoscale

Deploy and Configure a Virtual Machine Scale Set (VMSS) with Autoscale

STEP 1 OF 8

1

Create the resource group

```
$ az group create --name rg-az104-lab15 --location eastus
```

Deploy and Configure a Virtual Machine Scale Set (VMSS) with Autoscale

STEP 2 OF 8

2

Create the scale set behind a load balancer

```
$ az vmss create --resource-group rg-az104-lab15 --name vmss-lab15 --orchestration-mode Flexible --image Ubuntu2204 --vm-sku Standard_B1s --instance-count 2 --admin-username azureuser --generate-ssh-keys --load-balancer vmss-lb --upgrade-policy-mode automatic
```

Deploy and Configure a Virtual Machine Scale Set (VMSS) with Autoscale

STEP 3 OF 8

3

View and manually scale instances

```
$ az vmss scale --resource-group rg-az104-lab15 --name vmss-lab15 --new-capacity 3
```

Deploy and Configure a Virtual Machine Scale Set (VMSS) with Autoscale

STEP 4 OF 8

4

Create an autoscale profile with min/max/default

```
$ az monitor autoscale create --resource-group rg-az104-lab15 --resource vmss-lab15 --resource-type  
Microsoft.Compute/virtualMachineScaleSets --name autoscale-lab15 --min-count 2 --max-count 10 --count 2
```

Deploy and Configure a Virtual Machine Scale Set (VMSS) with Autoscale

STEP 5 OF 8

5

Add a scale-out rule when CPU is high

```
$ az monitor autoscale rule create --resource-group rg-az104-lab15 --autoscale-name autoscale-lab15 --condition "Percentage CPU > 70 avg 5m" --scale out 1
```


Deploy and Configure a Virtual Machine Scale Set (VMSS) with Autoscale

STEP 6 OF 8

6

Add a scale-in rule when CPU is low

```
$ az monitor autoscale rule create --resource-group rg-az104-lab15 --autoscale-name autoscale-lab15 --condition "Percentage CPU < 30 avg 5m" --scale in 1
```

Deploy and Configure a Virtual Machine Scale Set (VMSS) with Autoscale

STEP 7 OF 8

7

Review the autoscale configuration

```
$ az monitor autoscale rule list --resource-group rg-az104-lab15 --autoscale-name autoscale-lab15 --output table
```

Deploy and Configure a Virtual Machine Scale Set (VMSS) with Autoscale

STEP 8 OF 8

8

Update the instance SKU and roll to instances

```
$ az vmss update --resource-group rg-az104-lab15 --name vmss-lab15 --set sku.name=Standard_B2s
```

Deploy and Configure a Virtual Machine Scale Set (VMSS) with Autoscale

Test it

Confirm both scale rules and the min/max bounds appear in the autoscale settings.

Containers: ACR, Container Instances, and Container Apps

LAB 16

Build and push an image to Azure Container Registry, run it in Container Instances, then deploy and scale it in Container Apps.

You'll build

An image in ACR running in both Container Instances and a scalable Container App.

Azure services: Azure Container Registry, Azure Container Instances, Azure Container Apps, Log Analytics

Containers: ACR, Container Instances, and Container Apps

STEP 1 OF 8

1

Create the resource group

```
$ az group create --name rg-az104-lab16 --location eastus
```

Containers: ACR, Container Instances, and Container Apps

STEP 2 OF 8

2

Create an Azure Container Registry

```
$ az acr create --resource-group rg-az104-lab16 --name $ACR --sku Basic --admin-enabled true
```

Containers: ACR, Container Instances, and Container Apps

STEP 3 OF 8



3

Build and push an image with ACR Tasks

```
$ az acr build --registry $ACR --image webapp:v1 https://github.com/Azure-Samples/acr-build-helloworld-node.git
```


Containers: ACR, Container Instances, and Container Apps

STEP 4 OF 8

4

Provision a container with Container Instances

```
$ az container create --resource-group rg-az104-lab16 --name aci-lab16 --image $ACR_SERVER/webapp:v1 --registry-login-server $ACR_SERVER --registry-username $ACR_USER --registry-password $ACR_PASS --dns-name-label acilab16$RANDOM --ports 80 --cpu 1 --memory 1.5
```

Containers: ACR, Container Instances, and Container Apps

STEP 5 OF 8

5

Create a Container Apps environment

```
$ az containerapp env create --name aca-env-lab16 --resource-group rg-az104-lab16 --location eastus
```

Containers: ACR, Container Instances, and Container Apps

STEP 6 OF 8

6

Provision a container with Container Apps

```
$ az containerapp create --name aca-lab16 --resource-group rg-az104-lab16 --environment aca-env-lab16 --image  
$ACR_SERVER/webapp:v1 --registry-server $ACR_SERVER --registry-username $ACR_USER --registry-password  
$ACR_PASS --target-port 80 --ingress external --cpu 0.5 --memory 1.0Gi
```

Containers: ACR, Container Instances, and Container Apps

STEP 7 OF 8



Manage Container App sizing and scaling

```
$ az containerapp update --name aca-lab16 --resource-group rg-az104-lab16 --min-replicas 0 --max-replicas 5 --scale-rule-name http-rule --scale-rule-type http --scale-rule-http-concurrency 50
```

Containers: ACR, Container Instances, and Container Apps

STEP 8 OF 8



Manage registry images and usage

```
$ az acr repository delete --name $ACR --image webapp:v1 --yes
```

Containers: ACR, Container Instances, and Container Apps

Test it

Browse the ACI and Container App FQDNs and confirm the web app responds.

Azure App Service: Plans, Scaling, TLS, Custom Domains, Backup, Networking, Slots

LAB 17

Create an App Service plan and web app, then configure scaling, TLS, custom domains, backup, networking, and deployment slots.

You'll build

A Standard-tier web app with autoscale, TLS, backup, VNet integration, and a staging slot.

Azure services: Azure App Service, App Service plans, Azure Monitor Autoscale, Managed Certificates, VNet integration

Azure App Service: Plans, Scaling, TLS, Custom Domains, Backup, Networking, Slots

STEP 1 OF 9

1

Create the resource group

```
$ az group create --name rg-az104-lab17 --location eastus
```


Azure App Service: Plans, Scaling, TLS, Custom Domains, Backup, Networking, Slots

STEP 2 OF 9

2

Provision a Standard App Service plan

```
$ az appservice plan create --resource-group rg-az104-lab17 --name plan-lab17 --sku S1 --is-linux
```

Azure App Service: Plans, Scaling, TLS, Custom Domains, Backup, Networking, Slots

STEP 3 OF 9

3

Create an App Service web app

```
$ az webapp create --resource-group rg-az104-lab17 --plan plan-lab17 --name $APP --runtime "NODE:20-lts"
```

Azure App Service: Plans, Scaling, TLS, Custom Domains, Backup, Networking, Slots

STEP 4 OF 9

4

Configure scale up, scale out, and autoscale

```
$ az monitor autoscale create -g rg-az104-lab17 --resource $PLANID --name autoscale-lab17 --min-count 1 --max-count 5 --count 1
```

Azure App Service: Plans, Scaling, TLS, Custom Domains, Backup, Networking, Slots

STEP 5 OF 9

5

Map a custom DNS name

```
$ az webapp config hostname add --resource-group rg-az104-lab17 --webapp-name $APP --hostname  
www.contoso.com
```

Azure App Service: Plans, Scaling, TLS, Custom Domains, Backup, Networking, Slots

STEP 6 OF 9

6

Create and bind a managed TLS certificate

```
$ az webapp config ssl bind --resource-group rg-az104-lab17 --name $APP --certificate-thumbprint $THUMB --ssl-type SNI
```

Azure App Service: Plans, Scaling, TLS, Custom Domains, Backup, Networking, Slots

STEP 7 OF 9

7

Configure scheduled backups to storage

```
$ az webapp config backup update --resource-group rg-az104-lab17 --webapp-name $APP --container-url "$CONTAINER_URL" --frequency 1d --retain-one true --retention 30
```

Azure App Service: Plans, Scaling, TLS, Custom Domains, Backup, Networking, Slots

STEP 8 OF 9

8

Configure VNet integration and access restrictions

```
$ az webapp vnet-integration add -g rg-az104-lab17 -n $APP --vnet vnet-lab17 --subnet appsvc-subnet
```

Azure App Service: Plans, Scaling, TLS, Custom Domains, Backup, Networking, Slots

STEP 9 OF 9

9

Create a deployment slot and swap

```
$ az webapp deployment slot swap --resource-group rg-az104-lab17 --name $APP --slot staging --target-slot production
```


Azure App Service: Plans, Scaling, TLS, Custom Domains, Backup, Networking, Slots

Test it

Confirm the app serves over HTTPS with TLS 1.2 and the staging slot swaps into production.

Recap — Deploy and Manage Azure Compute Resources

- You can now: Automate deployment of resources by using ARM templates or Bicep files
- You can now: Create and configure virtual machines
- You can now: Manage virtual machines
- You can now: Deploy and configure a Virtual Machine Scale Set
- You can now: Create and configure containers
- You can now: Create and configure Azure App Service

TOPIC 04

Implement and Manage Virtual Networking

VNets · NSGs & ASGs · Bastion · Endpoints · DNS · Load Balancing

Key Concepts — Implement and Manage Virtual Networking

1

A virtual network (VNet) is your private network in Azure, divided into subnets; peering joins VNets.

2

NSGs and ASGs filter traffic; effective security rules show what actually applies to a NIC.

3

Azure Bastion gives browser RDP/SSH with no public IP; service/private endpoints lock PaaS to the VNet.

4

Azure DNS resolves names; load balancers distribute traffic across a backend pool with health probes.

Hands-On Labs — Implement and Manage Virtual Networking

Labs 18–19

- Virtual Networks, Subnets & VNet Peering
- Public IPs, User-Defined Routes & Connectivity Troubleshooting

Labs 20–21

- Network Security Groups, Application Security Groups & Effective Rules
- Azure Bastion, Service Endpoints & Private Endpoints

Labs 22–22

- Azure DNS & Load Balancing

Virtual Networks, Subnets & VNet Peering

LAB 18

Create two virtual networks with non-overlapping address spaces and subnets, then peer them so resources communicate over the Azure backbone without a VPN or public IPs.

You'll build

Two peered VNets (hub and spoke) in Connected state.

Azure services: Virtual Network, Subnets, VNet Peering, Network Watcher, az network vnet

Virtual Networks, Subnets & VNet Peering

STEP 1 OF 8



Create the resource group

```
$ az group create --name rg-az104-lab18 --location eastus
```

Virtual Networks, Subnets & VNet Peering

STEP 2 OF 8

2

Create first VNet vnet-hub with a subnet

```
$ az network vnet create --resource-group rg-az104-lab18 --name vnet-hub --address-prefixes 10.10.0.0/16 --subnet-name snet-hub-app --subnet-prefixes 10.10.1.0/24 --location eastus
```


Virtual Networks, Subnets & VNet Peering

STEP 3 OF 8

3

Create second VNet vnet-spoke with a subnet

```
$ az network vnet create --resource-group rg-az104-lab18 --name vnet-spoke --address-prefixes 10.20.0.0/16 --  
subnet-name snet-spoke-app --subnet-prefixes 10.20.1.0/24 --location eastus
```

Virtual Networks, Subnets & VNet Peering

STEP 4 OF 8

4

Add a second subnet to the hub VNet

```
$ az network vnet subnet create --resource-group rg-az104-lab18 --vnet-name vnet-hub --name snet-hub-mgmt --address-prefixes 10.10.2.0/24
```

Virtual Networks, Subnets & VNet Peering

STEP 5 OF 8

5

Create the peering from hub to spoke

```
$ az network vnet peering create --resource-group rg-az104-lab18 --name hub-to-spoke --vnet-name vnet-hub --remote-vnet vnet-spoke --allow-vnet-access
```

Virtual Networks, Subnets & VNet Peering

STEP 6 OF 8

6

Create the reverse peering from spoke to hub

```
$ az network vnet peering create --resource-group rg-az104-lab18 --name spoke-to-hub --vnet-name vnet-spoke --remote-vnet vnet-hub --allow-vnet-access
```

Virtual Networks, Subnets & VNet Peering

STEP 7 OF 8



Verify the peering state is Connected

```
$ az network vnet peering show --resource-group rg-az104-lab18 --vnet-name vnet-hub --name hub-to-spoke --query "{name:name, state:peeringState}" -o table
```

Virtual Networks, Subnets & VNet Peering

STEP 8 OF 8



8

Deploy test VMs and check connectivity

```
$ az network watcher test-connectivity --resource-group rg-az104-lab18 --source-resource vm-hub --dest-resource vm-spoke --dest-port 22
```

Virtual Networks, Subnets & VNet Peering

Test it

Both peerings show peeringState = Connected and test-connectivity across the peering returns Reachable.

Public IPs, User-Defined Routes & Connectivity Troubleshooting

LAB 19

Allocate a static Standard public IP, build a route table with a user-defined route forcing traffic through a virtual appliance, associate it to a subnet, then troubleshoot the resulting effective routes.

You'll build

A route table with a 0.0.0.0/0 UDR associated to a workload subnet.

Azure services: Public IP address, Route Table (UDR), Virtual Network, Network Watcher, NIC, az network route-table

Public IPs, User-Defined Routes & Connectivity Troubleshooting

STEP 1 OF 9

1

Create the resource group and a VNet

```
$ az network vnet create --resource-group rg-az104-lab19 --name vnet-lab19 --address-prefixes 10.30.0.0/16 --  
subnet-name snet-workload --subnet-prefixes 10.30.1.0/24
```

Public IPs, User-Defined Routes & Connectivity Troubleshooting

STEP 2 OF 9

2

Create a static Standard SKU public IP

```
$ az network public-ip create --resource-group rg-az104-lab19 --name pip-web-static --sku Standard --allocation-method Static --version IPv4
```

Public IPs, User-Defined Routes & Connectivity Troubleshooting

STEP 3 OF 9

3

Inspect the allocated address

```
$ az network public-ip show --resource-group rg-az104-lab19 --name pip-web-static --query "{ip:ipAddress, sku:sku.name}" -o table
```

Public IPs, User-Defined Routes & Connectivity Troubleshooting

STEP 4 OF 9

4

Create a route table

```
$ az network route-table create --resource-group rg-az104-lab19 --name rt-workload --location eastus
```

Public IPs, User-Defined Routes & Connectivity Troubleshooting

STEP 5 OF 9

5

Add a UDR to a virtual appliance

```
$ az network route-table route create --resource-group rg-az104-lab19 --route-table-name rt-workload --name route-to-nva --address-prefix 0.0.0.0/0 --next-hop-type VirtualAppliance --next-hop-ip-address 10.30.1.4
```

Public IPs, User-Defined Routes & Connectivity Troubleshooting

STEP 6 OF 9

6

Associate the route table to the subnet

```
$ az network vnet subnet update --resource-group rg-az104-lab19 --vnet-name vnet-lab19 --name snet-workload --route-table rt-workload
```

Public IPs, User-Defined Routes & Connectivity Troubleshooting

STEP 7 OF 9

7

Deploy a test VM with the static public IP

```
$ az vm create --resource-group rg-az104-lab19 --name vm-web --image Ubuntu2204 --size Standard_B1s --vnet-name vnet-lab19 --subnet snet-workload --public-ip-address pip-web-static --admin-username azureuser --generate-ssh-keys
```

Public IPs, User-Defined Routes & Connectivity Troubleshooting

STEP 8 OF 9

8

View the effective routes on the NIC

```
$ az network nic show-effective-route-table --resource-group rg-az104-lab19 --name "$NIC" -o table
```


Public IPs, User-Defined Routes & Connectivity Troubleshooting

STEP 9 OF 9

9

Troubleshoot connectivity with Network Watcher

```
$ az network watcher test-connectivity --resource-group rg-az104-lab19 --source-resource vm-web --dest-address 13.107.42.14 --dest-port 443
```

Public IPs, User-Defined Routes & Connectivity Troubleshooting

Test it

The NIC effective routes show a User-source 0.0.0.0/0 route to VirtualAppliance, and test-connectivity returns Unreachable confirming the UDR diverts traffic.

Network Security Groups, Application Security Groups & Effective Rules

LAB 20

Create an NSG with prioritized custom rules, group VMs by role using application security groups, reference those ASGs in NSG rules, then read the effective security rules Azure computes for a NIC.

You'll build

An NSG with ASG-based allow/deny rules associated to a subnet.

Azure services: Network Security Group, Application Security Group, Virtual Network, NIC, Network Watcher, az network nsg

Network Security Groups, Application Security Groups & Effective Rules

STEP 1 OF 9

1

Create the resource group, VNet and subnet

```
$ az network vnet create --resource-group rg-az104-lab20 --name vnet-lab20 --address-prefixes 10.40.0.0/16 --  
subnet-name snet-app --subnet-prefixes 10.40.1.0/24
```

Network Security Groups, Application Security Groups & Effective Rules

STEP 2 OF 9

2

Create application security groups

```
$ az network asg create --resource-group rg-az104-lab20 --name asg-web --location eastus
```

Network Security Groups, Application Security Groups & Effective Rules

STEP 3 OF 9

3

Create the network security group

```
$ az network nsg create --resource-group rg-az104-lab20 --name nsg-app --location eastus
```

Network Security Groups, Application Security Groups & Effective Rules

STEP 4 OF 9

4

Allow HTTPS from internet to the web ASG

```
$ az network nsg rule create --resource-group rg-az104-lab20 --nsg-name nsg-app --name Allow-HTTPS-Web --priority 100 --direction Inbound --access Allow --protocol Tcp --source-address-prefixes Internet --destination-asgs asg-web --destination-port-ranges 443
```

Network Security Groups, Application Security Groups & Effective Rules

STEP 5 OF 9

5

Allow SQL only from web ASG to db ASG

```
$ az network nsg rule create --resource-group rg-az104-lab20 --nsg-name nsg-app --name Allow-SQL-Web-to-DB --  
priority 110 --direction Inbound --access Allow --protocol Tcp --source-asgs asg-web --destination-asgs asg-db --  
destination-port-ranges 1433
```


Network Security Groups, Application Security Groups & Effective Rules

STEP 6 OF 9

6

Explicitly deny everything else inbound

```
$ az network nsg rule create --resource-group rg-az104-lab20 --nsg-name nsg-app --name Deny-All-Inbound --priority 4000 --direction Inbound --access Deny --protocol '*' --source-address-prefixes '*' --destination-address-prefixes '*' --destination-port-ranges '*'
```

Network Security Groups, Application Security Groups & Effective Rules

STEP 7 OF 9

7

Associate the NSG to the subnet

```
$ az network vnet subnet update --resource-group rg-az104-lab20 --vnet-name vnet-lab20 --name snet-app --network-security-group nsg-app
```

Network Security Groups, Application Security Groups & Effective Rules

STEP 8 OF 9

8

Deploy VMs and place NICs in the ASGs

```
$ az network nic ip-config update -g rg-az104-lab20 --nic-name "$WEB_NIC" -n ipconfig1 --application-security-groups asg-web
```

Network Security Groups, Application Security Groups & Effective Rules

STEP 9 OF 9

9

Evaluate the effective security rules

```
$ az network nic list-effective-nsg --resource-group rg-az104-lab20 --name "$WEB_NIC" -o table
```

Network Security Groups, Application Security Groups & Effective Rules

Test it

IP flow verify on vm-web returns Allow by Allow-HTTPS-Web for port 443 and Deny for port 22.

Azure Bastion, Service Endpoints & Private Endpoints

LAB 21

Deploy Azure Bastion for browser-based RDP/SSH without VM public IPs, lock a storage account to a subnet with a service endpoint, then bring that storage into the VNet with a private endpoint and private DNS.

You'll build

Azure Bastion plus a storage account reachable via a private endpoint.

Azure services: Azure Bastion, Virtual Network, Public IP, Service Endpoints, Private Endpoint, Private DNS Zone, Azure Storage

Azure Bastion, Service Endpoints & Private Endpoints

STEP 1 OF 9

1

Create the resource group and VNet

```
$ az network vnet create --resource-group rg-az104-lab21 --name vnet-lab21 --address-prefixes 10.50.0.0/16 --  
subnet-name snet-workload --subnet-prefixes 10.50.1.0/24
```

Azure Bastion, Service Endpoints & Private Endpoints

STEP 2 OF 9



2

Add the AzureBastionSubnet (/26 or larger)

```
$ az network vnet subnet create --resource-group rg-az104-lab21 --vnet-name vnet-lab21 --name AzureBastionSubnet  
--address-prefixes 10.50.2.0/26
```


Azure Bastion, Service Endpoints & Private Endpoints

STEP 3 OF 9



3

Create a Standard public IP for Bastion

```
$ az network public-ip create --resource-group rg-az104-lab21 --name pip-bastion --sku Standard --allocation-method Static
```

Azure Bastion, Service Endpoints & Private Endpoints

STEP 4 OF 9

4

Deploy Azure Bastion

```
$ az network bastion create --resource-group rg-az104-lab21 --name bastion-lab21 --vnet-name vnet-lab21 --public-ip-address pip-bastion --location eastus --sku Standard
```

Azure Bastion, Service Endpoints & Private Endpoints

STEP 5 OF 9

5

Deploy a VM with no public IP for Bastion

```
$ az vm create --resource-group rg-az104-lab21 --name vm-jump --image Ubuntu2204 --size Standard_B1s --vnet-name vnet-lab21 --subnet snet-workload --public-ip-address "" --admin-username azureuser --generate-ssh-keys
```

Azure Bastion, Service Endpoints & Private Endpoints

STEP 6 OF 9

6

Create a storage account for endpoint tests

```
$ az storage account create --resource-group rg-az104-lab21 --name $STG --sku Standard_LRS --kind StorageV2 --location eastus
```

Azure Bastion, Service Endpoints & Private Endpoints

STEP 7 OF 9

7

Enable a service endpoint on the subnet

```
$ az network vnet subnet update --resource-group rg-az104-lab21 --vnet-name vnet-lab21 --name snet-workload --service-endpoints Microsoft.Storage
```

Azure Bastion, Service Endpoints & Private Endpoints

STEP 8 OF 9



8

Restrict the storage account to that subnet

```
$ az storage account network-rule add --resource-group rg-az104-lab21 --account-name $STG --vnet-name vnet-lab21 --subnet subnet-workload
```

Azure Bastion, Service Endpoints & Private Endpoints

STEP 9 OF 9

9

Create a private endpoint and integrate DNS

```
$ az network private-endpoint create --resource-group rg-az104-lab21 --name pe-storage-blob --vnet-name vnet-lab21  
--subnet snet-workload --private-connection-resource-id "$STG_ID" --group-id blob --connection-name pe-storage-  
conn
```

Azure Bastion, Service Endpoints & Private Endpoints

Test it

From vm-jump via Bastion, nslookup of the storage blob endpoint resolves to a 10.50.x.x private IP instead of a public address.

Azure DNS & Load Balancing

LAB 22

Host a public DNS zone with A and CNAME records, then build a public Standard load balancer with a backend pool, health probe and rule across two VMs, and troubleshoot uneven traffic distribution.

You'll build

A public DNS zone plus a Standard load balancer distributing across two VMs.

Azure services: Azure DNS, Azure Load Balancer (Standard), Public IP, Virtual Network, Backend Pool, Health Probe, Network Watcher

Azure DNS & Load Balancing

STEP 1 OF 9



Create the resource group and VNet

```
$ az network vnet create --resource-group rg-az104-lab22 --name vnet-lab22 --address-prefixes 10.60.0.0/16 --  
subnet-name snet-backend --subnet-prefixes 10.60.1.0/24
```

Azure DNS & Load Balancing

STEP 2 OF 9

2

Create a public Azure DNS zone

```
$ az network dns zone create --resource-group rg-az104-lab22 --name contoso-az104-demo.com
```

Azure DNS & Load Balancing

STEP 3 OF 9

3

Add A and CNAME records to the zone

```
$ az network dns record-set a add-record --resource-group rg-az104-lab22 --zone-name contoso-az104-demo.com --record-set-name www --ipv4-address 20.1.2.3
```

Azure DNS & Load Balancing

STEP 4 OF 9

4

Create a Standard public IP for the LB

```
$ az network public-ip create --resource-group rg-az104-lab22 --name pip-lb --sku Standard --allocation-method Static
```

Azure DNS & Load Balancing

STEP 5 OF 9

5

Create the public load balancer with frontend

```
$ az network lb create --resource-group rg-az104-lab22 --name lb-web --sku Standard --public-ip-address pip-lb --  
frontend-ip-name fe-web --backend-pool-name pool-web
```

Azure DNS & Load Balancing

STEP 6 OF 9

6

Add a health probe

```
$ az network lb probe create --resource-group rg-az104-lab22 --lb-name lb-web --name probe-http --protocol Tcp --port 80 --interval 5
```

Azure DNS & Load Balancing

STEP 7 OF 9



Add a load-balancing rule

```
$ az network lb rule create --resource-group rg-az104-lab22 --lb-name lb-web --name rule-http --protocol Tcp --  
frontend-port 80 --backend-port 80 --frontend-ip-name fe-web --backend-pool-name pool-web --probe-name probe-http
```


Azure DNS & Load Balancing

STEP 8 OF 9



8

Deploy two backend VMs into the pool

```
$ az network nic ip-config address-pool add --resource-group rg-az104-lab22 --nic-name "$NIC" --ip-config-name ipconfig1 --lb-name lb-web --address-pool pool-web
```

Azure DNS & Load Balancing

STEP 9 OF 9

9

Test and troubleshoot load balancing

```
$ az monitor metrics list --resource $(az network lb show -g rg-az104-lab22 -n lb-web --query id -o tsv) --metric  
DipAvailability VipAvailability -o table
```

Azure DNS & Load Balancing

Test it

Browsing the LB frontend IP repeatedly alternates responses between vm-web1 and vm-web2, confirming traffic distribution.

Recap — Implement and Manage Virtual Networking

- You can now: Configure virtual networks
- You can now: Configure IP addressing and routing
- You can now: Secure network traffic
- You can now: Provide secure access to PaaS and VMs
- You can now: Configure name resolution and load balancing

TOPIC 05

Monitor and Maintain Azure Resources

Azure Monitor · Network Watcher · Backup · Site Recovery

Key Concepts — Monitor and Maintain Azure Resources

1

Azure Monitor collects metrics and logs (Log Analytics + KQL) and raises alerts via action groups.

2

Network Watcher diagnoses connectivity (IP flow verify, next hop, connection monitor).

3

Azure Backup protects data with point-in-time restore from a Recovery Services vault.

4

Azure Site Recovery replicates workloads to a second region for disaster-recovery failover.

Hands-On Labs — Monitor and Maintain Azure Resources

Labs 23–24

- Azure Monitor: Metrics, Logs, KQL, and Alerts
- Azure Network Watcher and Connection Monitor

Labs 25–26

- Azure Backup: Recovery Services Vault, Policies, Backup and Restore
- Azure Site Recovery: Replicate, Fail Over, and Interpret Health

—

- —

Azure Monitor: Metrics, Logs, KQL, and Alerts

LAB 23

Build the full Azure Monitor loop: a Log Analytics workspace, diagnostic settings, metrics, KQL queries, and alerts.

You'll build

A Log Analytics workspace with diagnostic data, KQL queries, and metric/activity/processing alert rules.

Azure services: Azure Monitor, Log Analytics workspace, Metrics, Alerts, Action Groups, Alert Processing Rules, VM Insights

Azure Monitor: Metrics, Logs, KQL, and Alerts

STEP 1 OF 9



Create resource group and Log Analytics workspace

```
$ az monitor log-analytics workspace create --resource-group "$RG" --workspace-name "$WS" --location "$LOCATION" --retention-time 30
```

Azure Monitor: Metrics, Logs, KQL, and Alerts

STEP 2 OF 9

2

Deploy a monitored VM and storage account

```
$ az vm create --resource-group "$RG" --name "vm-az104-lab23" --image Ubuntu2204 --size Standard_B1s --admin-username azureuser --generate-ssh-keys --public-ip-sku Standard
```

Azure Monitor: Metrics, Logs, KQL, and Alerts

STEP 3 OF 9

3

Read platform metrics such as Percentage CPU

```
$ az monitor metrics list --resource "$VM_ID" --metric "Percentage CPU" --aggregation Average --interval PT5M
```

Azure Monitor: Metrics, Logs, KQL, and Alerts

STEP 4 OF 9

4

Route logs via diagnostic settings to the workspace

```
$ az monitor diagnostic-settings create --name "diag-to-law" --resource "$STORAGE_ID/blobServices/default" --  
workspace "$WS_ID" --logs '[{"category":"StorageRead","enabled":true}]' --metrics  
'[{"category":"Transaction","enabled":true}]'
```

Azure Monitor: Metrics, Logs, KQL, and Alerts

STEP 5 OF 9

5

Enable VM Insights (Azure Monitor Agent + DCR)

```
$ az vm extension set --resource-group "$RG" --vm-name "vm-az104-lab23" --name AzureMonitorLinuxAgent --publisher Microsoft.Azure.Monitor --enable-auto-upgrade true
```

Azure Monitor: Metrics, Logs, KQL, and Alerts

STEP 6 OF 9



6

Query logs with KQL against Heartbeat/Perf/AzureActivity

```
$ az monitor log-analytics query --workspace "$WS_GUID" --analytics-query "Heartbeat | summarize LastHeartbeat = max(TimeGenerated) by Computer" --output table
```

Azure Monitor: Metrics, Logs, KQL, and Alerts

STEP 7 OF 9



Create an action group for notifications

```
$ az monitor action-group create --resource-group "$RG" --name "ag-az104-lab23" --short-name "az104ag" --action  
email admin angch@tertiaryinfotech.com
```

Azure Monitor: Metrics, Logs, KQL, and Alerts

STEP 8 OF 9

8

Create a metric alert on high CPU

```
$ az monitor metrics alert create --name "alert-vm-highcpu" --resource-group "$RG" --scopes "$VM_ID" --condition "avg Percentage CPU > 80" --window-size 5m --evaluation-frequency 1m --severity 3 --action "$AG_ID"
```


Azure Monitor: Metrics, Logs, KQL, and Alerts

STEP 9 OF 9

9

Create an alert processing rule across the RG

```
$ az monitor alert-processing-rule create --name "apr-add-ag" --resource-group "$RG" --scopes  
"/subscriptions/$SUB_ID/resourceGroups/$RG" --rule-type AddActionGroups --action-groups "$AG_ID"
```

Azure Monitor: Metrics, Logs, KQL, and Alerts

Test it

In Monitor → Logs the KQL queries return data and the metric/activity/processing alert rules appear under Monitor → Alerts.

Azure Network Watcher and Connection Monitor

LAB 24

Use Network Watcher's diagnostic tools to verify traffic rules, trace routing, capture packets, and continuously monitor reachability between VMs.

You'll build

Two VMs diagnosed with IP flow verify, next hop, NSG flow logs, packet capture, and a Connection Monitor.

Azure services: Azure Network Watcher, IP Flow Verify, Next Hop, NSG Diagnostics, Packet Capture, Connection Monitor, VNet, NSG, VMs

Azure Network Watcher and Connection Monitor

STEP 1 OF 9



Create resource group and virtual network

```
$ az network vnet create --resource-group "$RG" --name "vnet-az104-lab24" --address-prefix 10.10.0.0/16 --subnet-name "subnet-app" --subnet-prefix 10.10.1.0/24
```

Azure Network Watcher and Connection Monitor

STEP 2 OF 9

2

Deploy two VMs with an NSG allowing SSH/HTTP

```
$ az vm create -g "$RG" -n "vm-web" --image Ubuntu2204 --size Standard_B1s --vnet-name "vnet-az104-lab24" --subnet "subnet-app" --admin-username azureuser --generate-ssh-keys --public-ip-sku Standard --nsg "nsg-web"
```

Azure Network Watcher and Connection Monitor

STEP 3 OF 9

3

Run IP flow verify to check a 5-tuple

```
$ az network watcher test-ip-flow --vm "$VMWEB_ID" --direction Inbound --protocol TCP --local "10.10.1.4:80" --remote "13.68.0.1:50000"
```

Azure Network Watcher and Connection Monitor

STEP 4 OF 9

4

Trace routing with next hop

```
$ az network watcher show-next-hop --resource-group "$RG" --vm "vm-web" --source-ip 10.10.1.4 --dest-ip 8.8.8.8
```

Azure Network Watcher and Connection Monitor

STEP 5 OF 9

5

View effective NSG rules on the NIC

```
$ az network nic list-effective-nsg --ids "$NIC_ID" -o jsonc
```


Azure Network Watcher and Connection Monitor

STEP 6 OF 9

6

Enable NSG flow logs to a storage account

```
$ az network watcher flow-log create --location "$LOCATION" --name "flowlog-nsg-web" --nsg "$NSG_ID" --storage-account "$STORAGE_ID" --retention 7 --enabled true
```

Azure Network Watcher and Connection Monitor

STEP 7 OF 9



Capture packets on the VM for offline analysis

```
$ az network watcher packet-capture create --resource-group "$RG" --vm "vm-web" --name "pcap-vm-web" --storage-account "$STORAGE" --time-limit 60
```

Azure Network Watcher and Connection Monitor

STEP 8 OF 9



8

Run a point-in-time connection troubleshoot

```
$ az network watcher test-connectivity --resource-group "$RG" --source-resource "vm-web" --dest-address "$VMAPP_IP" --dest-port 22
```

Azure Network Watcher and Connection Monitor

STEP 9 OF 9

9

Configure a continuous Connection Monitor

```
$ az network watcher connection-monitor create --name "cm-web-to-app" --location "$LOCATION" --endpoint-source-name "src-vm-web" --endpoint-source-resource-id "$VMWEB_ID" --endpoint-dest-name "dst-vm-app" --endpoint-dest-address "$VMAPP_IP" --test-config-name "tcp-22" --protocol Tcp --tcp-port 22 --frequency 30
```

Azure Network Watcher and Connection Monitor

Test it

IP flow verify returns Allow/Deny with the deciding rule, and Connection Monitor shows ongoing reachability results between vm-web and vm-app.

Azure Backup: Recovery Services Vault, Policies, Backup and Restore

LAB 25

Create a Recovery Services vault, author a custom backup policy, protect a VM, run backups, and perform file-level and full VM restores.

You'll build

A Recovery Services vault protecting a VM with a custom policy, recovery points, and configured backup reports/alerts.

Azure services: Recovery Services vault, Azure Backup, Backup policy, Virtual Machine, Backup center / Backup reports, Azure Monitor alerts

Azure Backup: Recovery Services Vault, Policies, Backup and Restore

STEP 1 OF 9

1

Create resource group and Recovery Services vault

```
$ az backup vault create --resource-group "$RG" --name "$VAULT" --location "$LOCATION"
```

Azure Backup: Recovery Services Vault, Policies, Backup and Restore

STEP 2 OF 9

2

Set backup storage redundancy to GRS

```
$ az backup vault backup-properties set --resource-group "$RG" --name "$VAULT" --backup-storage-redundancy  
GeoRedundant
```


Azure Backup: Recovery Services Vault, Policies, Backup and Restore

STEP 3 OF 9

3

Deploy a VM to protect

```
$ az vm create --resource-group "$RG" --name "vm-az104-lab25" --image Ubuntu2204 --size Standard_B1s --admin-username azureuser --generate-ssh-keys --public-ip-sku Standard
```

Azure Backup: Recovery Services Vault, Policies, Backup and Restore

STEP 4 OF 9

4

Create a custom daily backup policy

```
$ az backup policy create --resource-group "$RG" --vault-name "$VAULT" --name "pol-daily-30d" --backup-management-type AzurelaasVM --policy @policy.json
```

Azure Backup: Recovery Services Vault, Policies, Backup and Restore

STEP 5 OF 9

5

Enable backup protection on the VM

```
$ az backup protection enable-for-vm --resource-group "$RG" --vault-name "$VAULT" --vm "vm-az104-lab25" --policy-name "pol-daily-30d"
```

Azure Backup: Recovery Services Vault, Policies, Backup and Restore

STEP 6 OF 9

6

Run an on-demand backup

```
$ az backup protection backup-now --resource-group "$RG" --vault-name "$VAULT" --container-name "vm-az104-lab25" --item-name "vm-az104-lab25" --backup-management-type AzurelaasVM --retain-until 31-07-2026
```

Azure Backup: Recovery Services Vault, Policies, Backup and Restore

STEP 7 OF 9

7

Restore individual files via ILR mount

```
$ az backup restore files mount-rp --resource-group "$RG" --vault-name "$VAULT" --container-name "vm-az104-lab25" --item-name "vm-az104-lab25" --rp-name "$RP"
```

Azure Backup: Recovery Services Vault, Policies, Backup and Restore

STEP 8 OF 9

8

Restore the full VM disks to staging

```
$ az backup restore restore-disks --resource-group "$RG" --vault-name "$VAULT" --container-name "vm-az104-lab25" --item-name "vm-az104-lab25" --rp-name "$RP" --storage-account "$STORAGE" --target-resource-group "$RG"
```

Azure Backup: Recovery Services Vault, Policies, Backup and Restore

STEP 9 OF 9

9

Configure vault diagnostics for backup reports/alerts

```
$ az monitor diagnostic-settings create --name "vault-diag" --resource "$VAULT_ID" --workspace "$WS_ID" --logs '[{"category":"CoreAzureBackup","enabled":true},{"category":"AddonAzureBackupJobs","enabled":true}]'
```

Azure Backup: Recovery Services Vault, Policies, Backup and Restore

Test it

az backup item list shows the VM protected with recovery points, and the restore jobs complete successfully in Backup Jobs.

Azure Site Recovery: Replicate, Fail Over, and Interpret Health

LAB 26

Enable Azure-to-Azure replication for a VM, run test and real failovers with re-protect and failback, and interpret replication health.

You'll build

A VM replicated to a secondary region via ASR with test/real failover, failback, and a Recovery Plan.

Azure services: Recovery Services vault (Site Recovery), Azure Site Recovery (Azure-to-Azure), source VM, source/target VNets, cache storage account

Azure Site Recovery: Replicate, Fail Over, and Interpret Health

STEP 1 OF 9

1

Create resource group, source VM, and vault

```
$ az backup vault create --resource-group "$RG" --name "$VAULT" --location "$PRIMARY"
```

Azure Site Recovery: Replicate, Fail Over, and Interpret Health

STEP 2 OF 9

2

Pre-create DR resource group, VNet, and cache storage

```
$ az network vnet create --resource-group "rg-az104-lab26-dr" --name "vnet-dr" --location "$SECONDARY" --address-prefix 10.20.0.0/16 --subnet-name "default" --subnet-prefix 10.20.1.0/24
```

Azure Site Recovery: Replicate, Fail Over, and Interpret Health

STEP 3 OF 9



3

Enable Site Recovery replication for the VM

Azure Site Recovery: Replicate, Fail Over, and Interpret Health

STEP 4 OF 9



4

Adjust the replication policy (snapshot frequency/retention)

Azure Site Recovery: Replicate, Fail Over, and Interpret Health

STEP 5 OF 9

5

Interpret replication health, status, and RPO

```
$ az resource list --resource-group "$RG" --resource-type "Microsoft.RecoveryServices/vaults" -o table
```

Azure Site Recovery: Replicate, Fail Over, and Interpret Health

STEP 6 OF 9



6

Run a non-disruptive test failover into an isolated VNet

Azure Site Recovery: Replicate, Fail Over, and Interpret Health

STEP 7 OF 9

7

Perform a failover to the secondary region and commit

```
$ az vm list --resource-group "rg-az104-lab26-dr" -o table
```


Azure Site Recovery: Replicate, Fail Over, and Interpret Health

STEP 8 OF 9



Re-protect and fail back to the primary region

Azure Site Recovery: Replicate, Fail Over, and Interpret Health

STEP 9 OF 9



Build a Recovery Plan for orchestrated failover

Azure Site Recovery: Replicate, Fail Over, and Interpret Health

Test it

The replicated item reaches Protected/Healthy with a current RPO, and a test failover creates a working VM in the secondary region.

Recap — Monitor and Maintain Azure Resources

- You can now: Monitor resources with Azure Monitor
- You can now: Monitor networks with Azure Network Watcher
- You can now: Implement backup and recovery
- You can now: Implement disaster recovery



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

Identities & Governance

Managed Entra identities, RBAC, Policy, resource groups, locks, tags and cost.

2

Storage

Configured storage accounts, redundancy, access control, Blob and Files.

3

Compute

Deployed with ARM/Bicep; ran VMs, scale sets, containers and App Service.

4

Virtual Networking

Built VNets, NSGs/ASGs, Bastion, endpoints, DNS and load balancing.

5

Monitor & Maintain

Used Azure Monitor, Network Watcher, Backup and Site Recovery.

Preparing for the AZ-104 Exam

- Redo every lab using only the Azure CLI until the commands are automatic.
- Review the 'What you learned' bullets in each lab and the Learner Guide.
- Take the free Microsoft practice assessment for AZ-104.
- Passing score is 700/1000; book the exam from your Microsoft Learn profile.
- Always run the clean-up step to keep your Azure bill near zero.

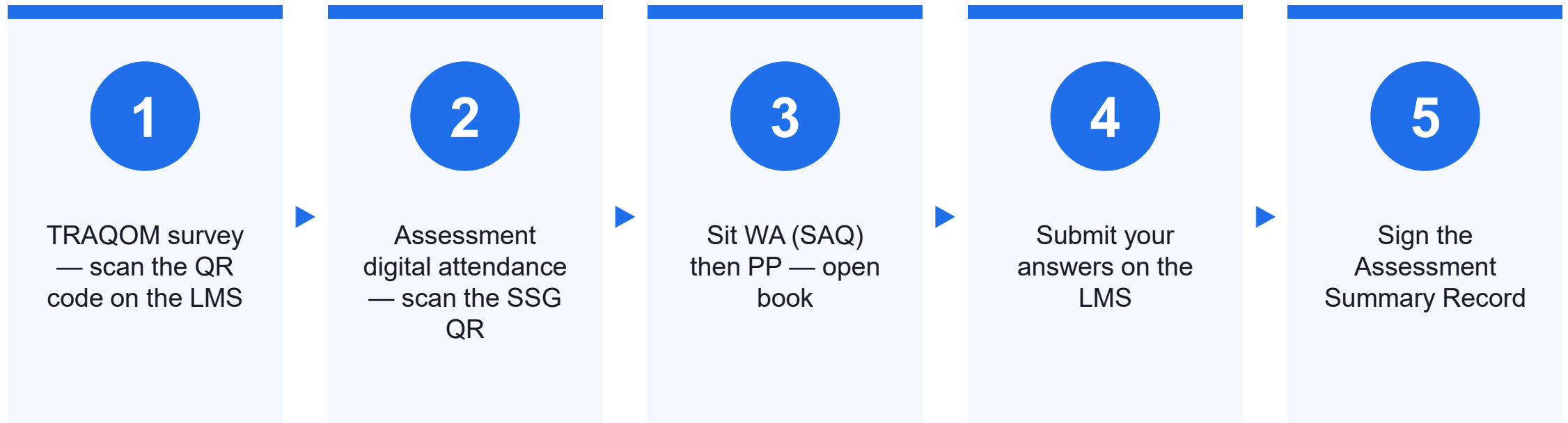
Practice Exam

- Sharpen your exam readiness with the Tertiary Infotech AZ-104 practice exam.
- Practice exam: <https://exams.tertiaryinfotech.com/practice-exams/microsoft/microsoft-az-104>
- Attempt it under timed conditions and review every explanation.
- Revisit any lab whose topic you miss, then re-take the practice exam.

Assessment

- Written Assessment (SAQ) — 1 hour. Practical Performance (PP) — 1 hour.
- Open book: slides, Learner Guide and approved materials only.
- Remember to take the Assessment digital attendance (TRAQOM).
- Submit your completed answers on the LMS at <https://lms-tms.tertiaryinfotech.com/>.

Assessment Flow



Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

SEE YOU IN THE CLOUD

Thank You!

You are now ready to administer Microsoft Azure — and to sit the AZ-104 exam.